

SAVEETHA SCHOOL OF ENGINEERING
TRANSMISSION LINES & WAVEGUIDES – ECA15
EXPERIMENT – 1

Estimation of Characteristic Impedance (Z_0) and Propagation Constant (γ) of a Transmission Line

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Date	07/10/2026	Experiment No.	1

1. AIM

To investigate the significance of characteristic impedance (Z_0) and propagation constant (γ) of a transmission line and study the impact of resistance, inductance, frequency and capacitance using SCILAB.

2. SOFTWARE REQUIRED

SCILAB Version 6.1.1

3. THEORY

$$Z_0 = \sqrt{[(R + j\omega L)/(G + j\omega C)]}$$

$$\gamma = \sqrt{[(R + j\omega L)(G + j\omega C)]} = \alpha + j\beta$$

where $\omega = 2\pi f$, α is the attenuation constant in neper/km and β is the phase constant in radians/km.

4. PROCEDURE

- Open SCILAB software.
- Open File → New.
- Enter the required input values.
- Type the program.
- Execute the program.
- Record the characteristic impedance, attenuation constant and phase constant.
- Repeat for each specified test case.

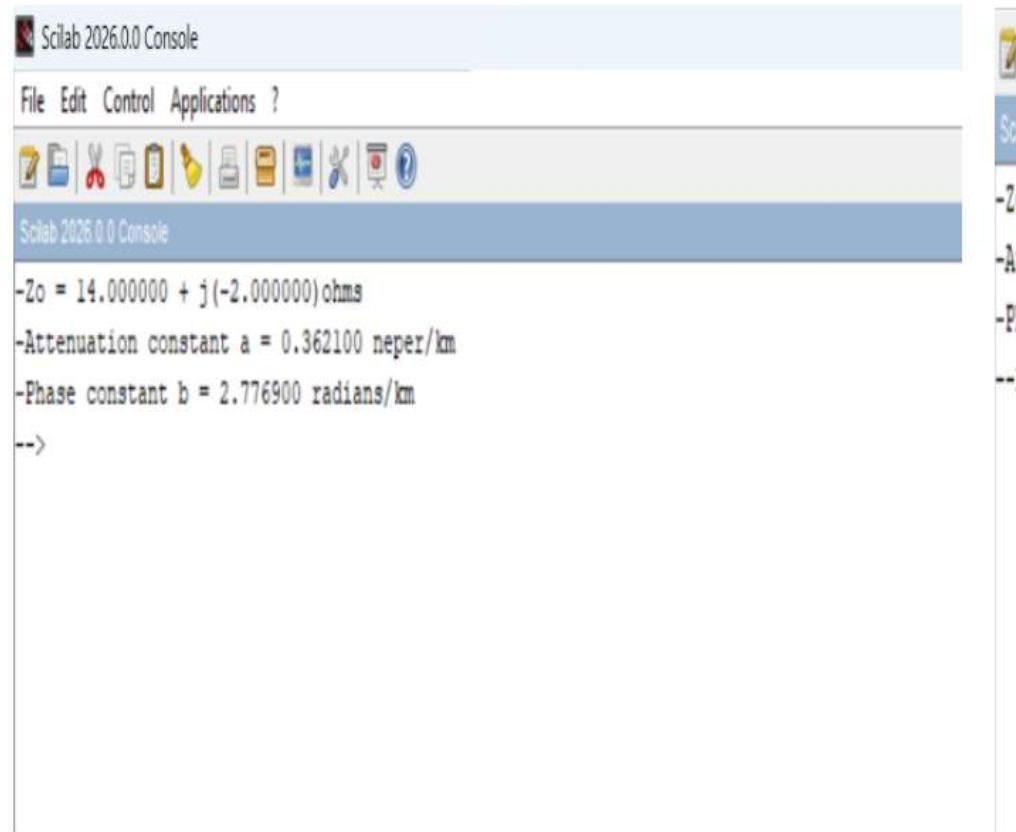
5. TEST CASE 1 – BASIC CALCULATION

Given in the manual: $R = 10 \Omega/\text{km}$, $C = 8 \mu\text{F}/\text{km}$, $L = 3.5 \text{ mH}/\text{km}$, $G = 4 \mu\text{mho}/\text{km}$, $f = 2 \text{ kHz}$.

The supplied SCILAB output file gives: $Z_0 = 14 - j2 \Omega$, $\alpha = 0.3621 \text{ neper}/\text{km}$, $\beta = 2.7769 \text{ radians}/\text{km}$.

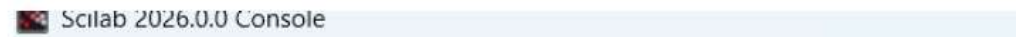
S.No.	Input condition	Simulation $Z_0 (\Omega)$	Simulation α	Simulation β	Theoretical α, β
1	$R=10, L=3.5 \text{ mH}, C=8 \mu\text{F}, G=4 \mu\text{mho}, f=2 \text{ kHz}$	$14 - j2$	0.3621	2.7769	See note below

Test case 1



```
Scilab 2026.0.0 Console
File Edit Control Applications ?
Scilab 2026.0.0 Console
-Zo = 14.000000 + j(-2.000000)ohms
-Attenuation constant a = 0.362100 neper/km
-Phase constant b = 2.776900 radians/km
-->
```

Test case 2



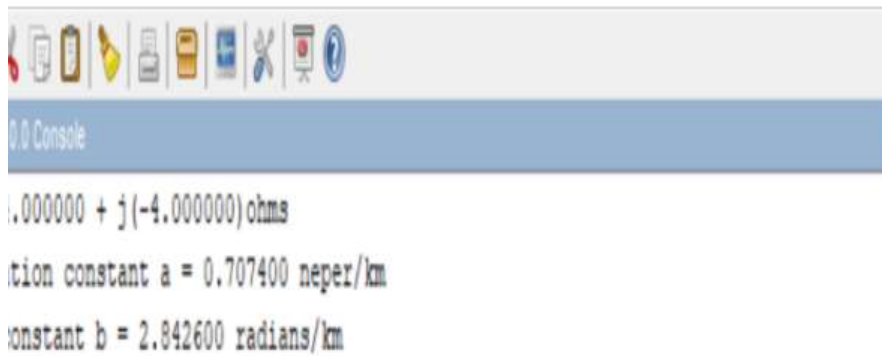
```
Scilab 2026.0.0 Console
```

6. TEST CASE 2A – VARIABLE RESISTANCE (R)

At $f = 2.5$ kHz; $L = 3.5$ mH/km, $G = 4$ μ mho/km and $C = 8$ μ F/km are kept constant. R is varied from 10 to 50 Ω /km in steps of 10 Ω /km.

S.No.	Inputs	Z_o (Ω)	α (neper/km)	β (radians/km)	α (neper/km)	β (radians/km)
1	R = 10 Ω /km	21.00 – j1.89	0.2381	2.6392	0.2381	2.6392
2	R = 20 Ω /km	21.25 – j3.74	0.4707	2.6702	0.4707	2.6702
3	R = 30 Ω /km	21.63 – j5.52	0.6935	2.7184	0.6935	2.7184
4	R = 40 Ω /km	22.12 – j7.19	0.9042	2.7796	0.9042	2.7796
5	R = 50 Ω /km	22.68 – j8.77	1.1023	2.8502	1.1023	2.8502

Test case 2



```

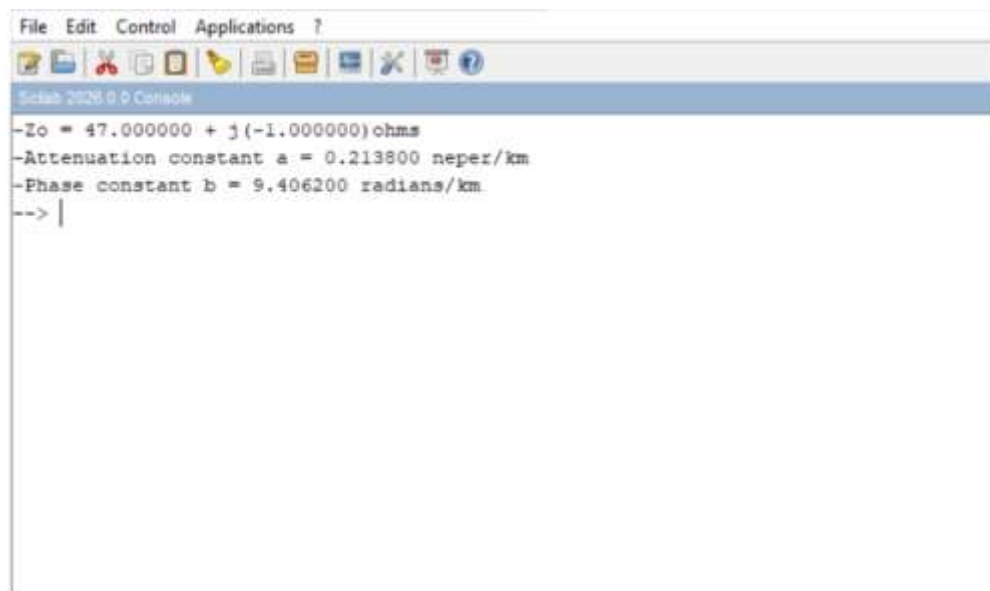
1.000000 + j(-4.000000) ohms
Attenuation constant a = 0.707400 neper/km
Phase constant b = 2.842600 radians/km

```

7. TEST CASE 2B – VARIABLE INDUCTANCE (L)

At $f = 2.5$ kHz; $R = 10 \Omega/\text{km}$, $G = 4 \mu\text{mho}/\text{km}$ and $C = 8 \mu\text{F}/\text{km}$ are kept constant. L is varied from 3.5 to 11.5 mH/km in steps of 2.5 mH/km, as stated in the manual.

S.No.	Inputs	$Z_0 (\Omega)$	α (neper/km)	β (radians/km)	α (neper/km)	β (radians/km)
1	$L = 3.5$ mH/km	$21.00 - j1.89$	0.2381	2.6392	0.2381	2.6392
2	$L = 6.0$ mH/km	$27.42 - j1.45$	0.1824	3.4463	0.1824	3.4463
3	$L = 8.5$ mH/km	$32.62 - j1.22$	0.1534	4.0990	0.1534	4.0990
4	$L = 11.0$ mH/km	$37.10 - j1.07$	0.1349	4.6617	0.1349	4.6617
5	$L = 11.5$ mH/km	$37.93 - j1.05$	0.1319	4.7663	0.1319	4.7663



```

File Edit Control Applications ?
-Zo = 47.000000 + j(-1.000000) ohms
-Attenuation constant a = 0.213800 neper/km
-Phase constant b = 9.406200 radians/km
--> |

```

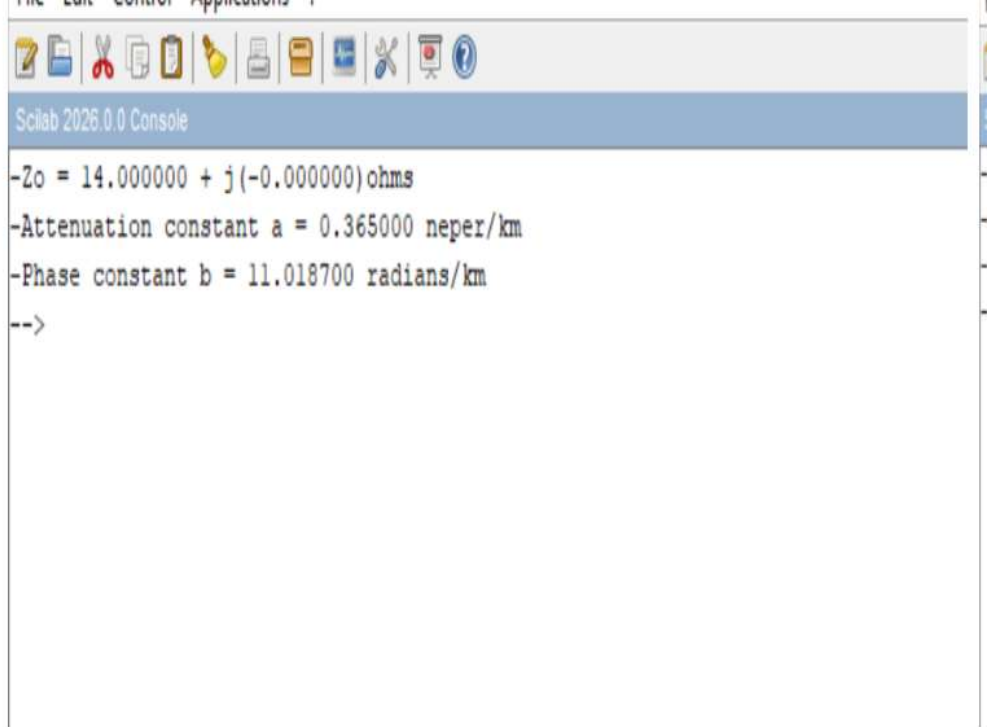
Test case 3



8. TEST CASE 3A – VARIABLE FREQUENCY (f)

$L = 3 \text{ mH/km}$, $R = 10 \text{ } \Omega/\text{km}$, $G = 4 \text{ } \mu\text{mho/km}$ and $C = 8 \text{ } \mu\text{F/km}$ are kept constant. Frequency is varied from 2 to 10 kHz in steps of 2 kHz.

S.No.	Inputs	$Z_o (\Omega)$	α (neper/km)	β (radians/km)	α (neper/km)	β (radians/km)
1	$f = 2 \text{ kHz}$	$19.53 - j2.55$	0.2560	1.9635	0.2560	1.9635
2	$f = 4 \text{ kHz}$	$19.41 - j1.28$	0.2577	3.9021	0.2577	3.9021
3	$f = 6 \text{ kHz}$	$19.38 - j0.86$	0.2580	5.8460	0.2580	5.8460
4	$f = 8 \text{ kHz}$	$19.38 - j0.64$	0.2581	7.7914	0.2581	7.7914
5	$f = 10 \text{ kHz}$	$19.37 - j0.51$	0.2581	9.7373	0.2581	9.7373

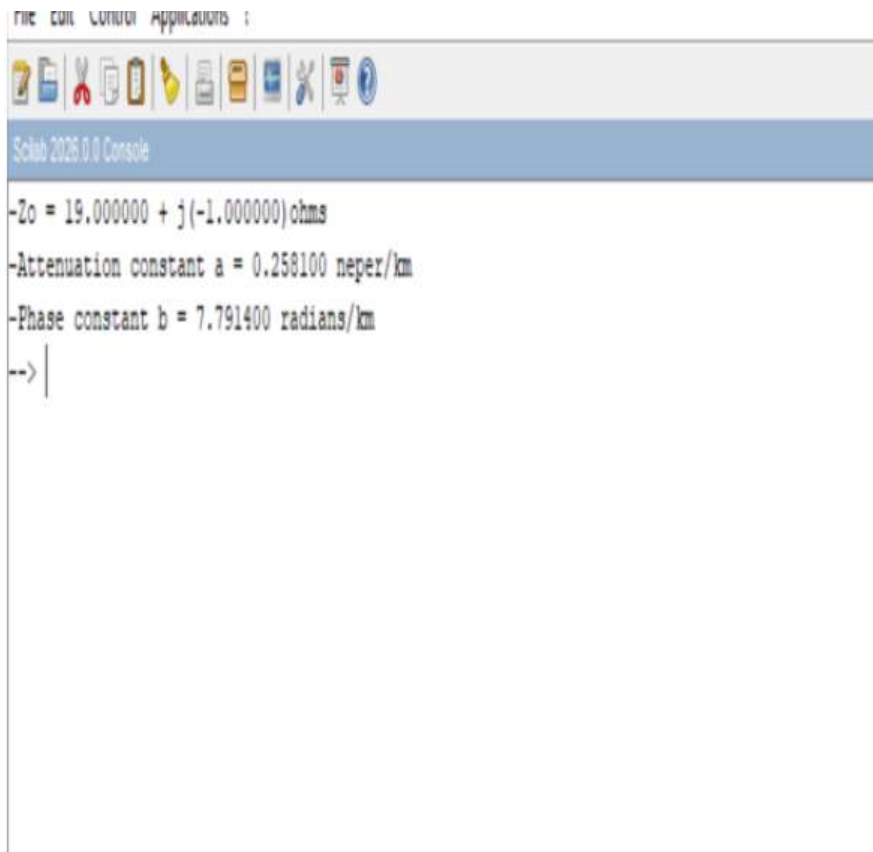


9. TEST CASE 3B – VARIABLE CAPACITANCE (C)

As printed on the manual page: $R = 10 \text{ } \Omega/\text{km}$, $L = 9 \text{ mH/km}$, $f = 2.5 \text{ kHz}$ and $G = 0.4 \text{ } \mu\text{mho/km}$. C is varied from 8 to 16 $\mu\text{F/km}$ in steps of 2 $\mu\text{F/km}$.

S.No.	Inputs	$Z_o (\Omega)$	α (neper/km)	β (radians/km)	α (neper/km)	β (radians/km)
1	$C = 8 \text{ } \mu\text{F/km}$	$33.56 - j1.19$	0.1490	4.2175	0.1490	4.2175
2	$C = 10 \text{ } \mu\text{F/km}$	$30.02 - j1.06$	0.1666	4.7153	0.1666	4.7153
3	$C = 12 \text{ } \mu\text{F/km}$	$27.40 - j0.97$	0.1825	5.1654	0.1825	5.1654
4	$C = 14 \text{ } \mu\text{F/km}$	$25.37 - j0.90$	0.1971	5.5793	0.1971	5.5793
5	$C = 16 \text{ } \mu\text{F/km}$	$23.73 - j0.84$	0.2107	5.9645	0.2107	5.9645

Observed SCILAB output supplied for this test case: $Z_o = 19 - j1 \text{ } \Omega$, $\alpha = 0.2581 \text{ neper/km}$, $\beta = 7.7914 \text{ rad/km}$. The screenshot does not show the corresponding capacitance value, so it is not assigned to a particular row.



10. SCILAB PROGRAM

```
clear; clc;
R=10; L=0.003; f=2*10^3; G=4*(10^-6); C=16*(10^-6);
w=2*pi*f;
Z=R+(%i*w*L);
Y=G+(%i*w*C);
Zo=sqrt(Z/Y);
C=round(real(Zo));
D=round(imag(Zo));
printf('-Zo = %f + j(%f)ohms\n',C,D);
P=sqrt(Z*Y);
a=real(P);
a1=round(a*10000)/10000;
printf('-Attenuation constant a = %f neper/km\n',a1);
b=imag(P);
b1=round(b*10000)/10000;
printf('-Phase constant b = %f radians/km',b1);
```

11. RESULT

The characteristic impedance and propagation constant of the transmission line were calculated using SCILAB. The supplied output file contains the observed console results for Test Cases 1, 2 and 3, and the five required tabular sections have been reproduced in the report.